AGILE PROJECT ARTIFACTS (SPRINT BACKLOG, REPORTS)

SAP BTP Integration Development

Employee Data Synchronization (SF to S/4HANA) using SAP Cloud Integration

CAPSTONE PROJECT

KRISHNAKANT PAL
PID: P2012786192

1. Agile Project Overview

The project follows the Agile Scrum methodology to plan, develop, test and document the employee data integration solution.
The work is divided into smaller tasks and managed through a Product Backlog and Sprint Backlog. Each sprint focuses on a specific set of development, configuration, testing and documentation activities.

The Agile process used for the project consists of:

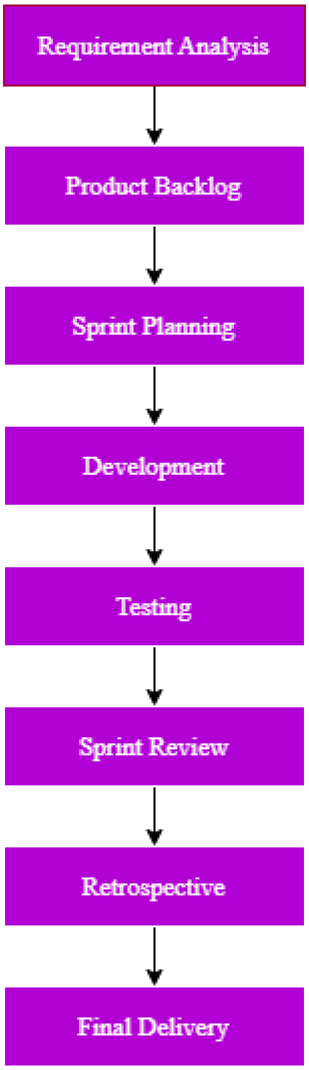

Figure 1: Agile Project Workflow Flowchart

The project follows the SDLC phases of Requirement Analysis, Design, Development, Testing, Deployment and Maintenance, as specified in the capstone requirements.

2. Project Goal

The goal of the project is to develop an integration solution that:

- Retrieves employee data from SAP SuccessFactors.
- Transforms the employee data.
- Sends the data to SAP S/4HANA.
- Generates a CSV file from the processed data.
- Uploads the CSV file to SFTP.

- Sends success notifications through email.
- Handles integration failures using an Exception Subprocess.
- Sends failure notifications through email.

3. Product Backlog

The Product Backlog contains the major requirements and activities identified for the project.

ID	Product Backlog Item	Priority	Status
PB-01	Analyze project requirements	High	Completed
PB-02	Design integration architecture	High	Completed
PB-03	Configure SuccessFactors connection	High	Completed
PB-04	Configure S/4HANA OData connection	High	Completed
PB-05	Create Message Mapping	High	Completed
PB-06	Configure XML to CSV conversion	Medium	Completed
PB-07	Configure SFTP upload	High	Completed
PB-08	Configure success email notification	Medium	Completed
PB-09	Implement Exception Subprocess	High	Completed
PB-10	Test successful integration	High	Completed
PB-11	Test SFTP failure scenario	High	Completed
PB-12	Test S/4HANA duplicate ID error	High	Completed
PB-13	Prepare technical documentation	Medium	Completed
PB-14	Prepare Agile artifacts	Medium	Completed
PB-15	Prepare presentation and demo	Medium	Completed
PB-16	Prepare final submission	High	Completed

4. Sprint Planning

The project work can be organized into two main development sprints.

Sprint 1 – Integration Development

Goal: Build the complete integration flow and configure all required system connections.

Planned activities:

- Requirement analysis
- Integration architecture
- SuccessFactors configuration
- S/4HANA OData configuration
- Message Mapping
- XML to CSV conversion
- SFTP configuration
- Mail configuration

Sprint 2 – Testing and Finalization

Goal: Validate the integration, implement error handling and prepare project documentation.

Planned activities:

- Exception Subprocess
- Successful end-to-end testing
- SFTP failure testing
- Duplicate Participant ID testing
- Success and failure email verification
- Technical documentation
- Agile documentation
- Presentation preparation
- Final submission preparation

5. Sprint Backlog – Sprint 1

Task ID	Task	Priority	Status
S1-01	Analyze capstone requirements	High	Completed
S1-02	Design iFlow architecture	High	Completed
S1-03	Configure SuccessFactors adapter	High	Completed
S1-04	Configure S/4HANA OData adapter	High	Completed
S1-05	Generate source and target XSDs	High	Completed
S1-06	Create Message Mapping	High	Completed
S1-07	Configure XML to CSV Converter	Medium	Completed
S1-08	Configure SFTP Receiver	High	Completed
S1-09	Configure Success Mail	Medium	Completed

Sprint 1 Goal

The complete basic integration flow was configured and the main system connections were established.

6. Sprint 1 Review

The Sprint 1 activities successfully established the core integration flow.

The following functionality was completed:

- SuccessFactors data retrieval
- S/4HANA OData integration
- Message Mapping
- XML to CSV conversion
- SFTP configuration
- Success email configuration

The basic end-to-end flow was ready for testing.

Sprint 1 Outcome

Sprint Status: Completed

7. Sprint 2 Backlog

Task ID	Task	Priority	Status
S2-01	Configure Exception Subprocess	High	Completed
S2-02	Configure failure email	High	Completed
S2-03	Test successful execution	High	Completed
S2-04	Verify SFTP CSV output	High	Completed
S2-05	Test SFTP failure	High	Completed
S2-06	Test duplicate Participant ID	High	Completed
S2-07	Verify failure email	High	Completed
S2-08	Prepare technical documentation	Medium	Completed
S2-09	Prepare Agile artifacts	Medium	Completed
S2-10	Prepare presentation and demo	Medium	Completed

8. Sprint 2 Review

During Sprint 2, the integration flow was tested under both successful and failure conditions.

The following scenarios were verified:

1. Successful employee data processing.
2. Successful SFTP file upload.
3. Successful email notification.
4. SFTP connection failure.
5. Duplicate Participant ID error from S/4HANA.
6. Exception Subprocess execution.
7. Failure email notification.

The testing confirmed that the integration flow can handle both normal and error scenarios.

Sprint 2 Outcome

Sprint Status: Completed

9. Sprint Retrospective

What Went Well

- ❖ The integration flow was successfully designed and deployed.
- ❖ SuccessFactors and S/4HANA connectivity was established.
- ❖ Message Mapping worked as expected.
- ❖ CSV generation and SFTP upload were successfully verified.
- ❖ Exception handling was tested with actual failure scenarios.
- ❖ Success and failure email notifications were verified.

Challenges Faced

- ❖ S/4HANA OData configuration required verification of the available entity and fields.
- ❖ Duplicate Participant ID resulted in a target-system error during testing.
- ❖ SFTP failure scenario had to be tested separately to verify Exception Subprocess behavior.

- ❖ Different system responses were analyzed through the Message Monitor.

Improvements

- ❖ Validate target-system constraints before repeated testing.
- ❖ Maintain separate positive and negative test cases.
- ❖ Capture monitoring evidence during each important test.
- ❖ Keep configuration and test documentation updated during development.

10. Sprint Progress Report

The project progress was tracked throughout the development and testing activities. The major integration tasks were completed progressively across the planned sprints.

Sprint Progress

Sprint	Planned Work	Completed Work	Status
Sprint 1	Integration design and development	Core iFlow, adapters, mapping, SFTP and Mail configured	Completed
Sprint 2	Testing, exception handling and documentation	Positive/negative testing, Exception Subprocess and documentation	Completed

Overall Progress

Requirement Analysis	100%
Architecture & Design	100%
Integration Development	100%
Testing	100%
Exception Handling	100%
Documentation	100%
Agile Artifacts	100%
Presentation & Demo	100%
Final Submission	100%

Note: The progress percentages above are project-status estimates for documentation purposes and can be updated as the remaining deliverables are completed.

11. Burndown Chart

A Burndown Chart can be used to represent the reduction of pending project tasks during the sprints.

Suggested Burndown Data

Sprint	Planned Tasks Remaining	Actual Tasks Remaining
Sprint Start	16	16
Sprint 1	8	7
Sprint 2	0	0

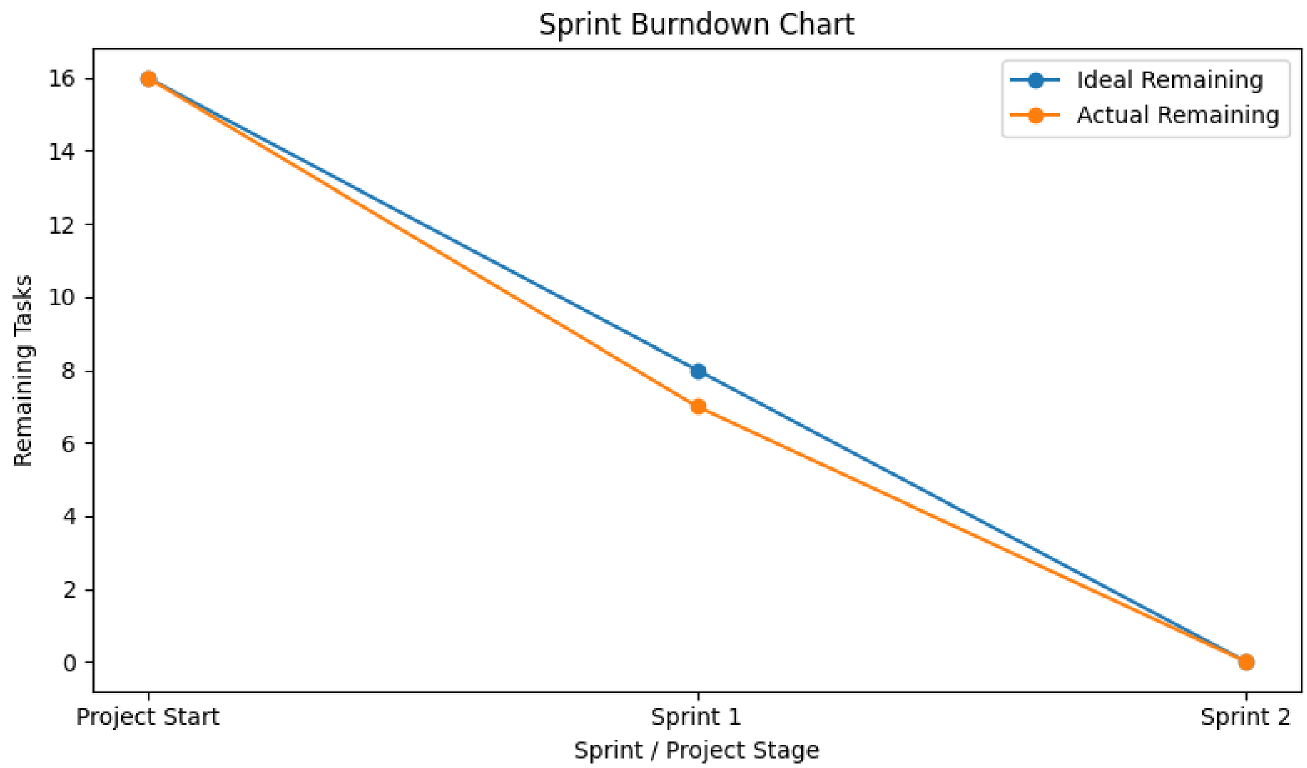

Figure 2: Sprint Burndown Chart

12. Sprint Review Summary

Sprint reviews were conducted to evaluate the completed work against the planned sprint objectives.

Sprint 1 Review

The first sprint focused on building the core integration solution.

Completed:

- ❖ SuccessFactors connectivity
- ❖ S/4HANA OData connectivity
- ❖ Message Mapping
- ❖ XML to CSV conversion
- ❖ SFTP configuration
- ❖ Success Mail configuration

Result: Sprint objective achieved.

Sprint 2 Review

The second sprint focused on testing and error handling.

Completed:

- ❖ Exception Subprocess

- Failure email
- Successful E2E test
- SFTP failure test
- Duplicate Participant ID test
- Test documentation

Result: Sprint objective achieved.

13. Agile Definition of Done

A task was considered Done when the required configuration was completed, tested and verified.

The following criteria were used:

- Configuration completed successfully.
- Required connectivity verified.
- Integration flow validated.
- Expected output generated.
- Error scenario tested where applicable.
- Relevant screenshots/evidence captured.
- Documentation updated.

14. Agile Project Metrics

The following project metrics summarize the implementation progress.

Metric	Result
Planned Sprints	2
Completed Sprints	2
Core Integration Flow	Completed
Positive Test Scenario	Passed
SFTP Failure Scenario	Passed
S/4HANA Duplicate ID Scenario	Passed
Exception Handling	Verified
Success Email	Verified
Failure Email	Verified

15. Final Agile Status

The planned development and testing activities were completed through the two sprints.

The integration solution was successfully deployed and tested using both positive and negative scenarios. The Exception Subprocess and email notification mechanism were also verified.

Final Status

- **Project Development:** Completed
- **Testing:** Completed
- **Exception Handling:** Completed
- **Technical Documentation:** Completed

- **Agile Documentation:** Completed
- **Presentation/Demo:** Pending Final Preparation
- **Final Submission:** Pending

16. Agile Conclusion

The Agile approach provided a structured way to divide the integration project into development and testing activities.

The first sprint focused on designing and implementing the integration flow, while the second sprint focused on testing, exception handling and project finalization.

The sprint reviews and retrospective helped identify implementation challenges and verify that the planned objectives were achieved.

17. Final Artifact Summary

The following Agile project artifacts were prepared and maintained during the project:

1. Product Backlog – Contains the complete list of project requirements and features.
2. Sprint Backlog – Sprint 1 – Contains the tasks planned and completed during Sprint 1.
3. Sprint Backlog – Sprint 2 – Contains the tasks planned and completed during Sprint 2.
4. Sprint 1 Review – Summarizes the work completed and outcomes of Sprint 1.
5. Sprint 2 Review – Summarizes the work completed and outcomes of Sprint 2.
6. Sprint Retrospective – Documents the challenges, improvements, and lessons learned during the sprints.
7. Sprint Progress Report – Provides the overall progress of the project across the sprints.
8. Burndown Chart – Shows the reduction of remaining tasks during the project.
9. Definition of Done – Defines the conditions used to consider a task successfully completed.
10. Agile Project Metrics – Summarizes project progress using task completion and sprint-related metrics.
11. Final Agile Status – Provides the final status of the Agile project and confirms completion of the planned activities.